

Untitled

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#Setup code

```
student_id <- c("S01", "S02", "S03", "S04", "S05", "S06")
section <- c("A", "B", "A", "B", "A", "B")
quiz1 <- c(82, 91, 76, 88, 95, 69)
quiz2 <- c(85, 89, 80, 92, 94, 74)
passed <- c(TRUE, TRUE, TRUE, TRUE, TRUE, FALSE)
```

Part A

```
section <- factor(section, levels = c("A", "B"))

# data frame containing all five vectors
students <- data.frame(
  student_id = student_id,
  section     = section,
  quiz1       = quiz1,
  quiz2       = quiz2,
  passed      = passed,
  stringsAsFactors = FALSE
)

# matrix: students as rows, quizzes as columns
score_matrix <- matrix(
  c(quiz1, quiz2),
  nrow = 6, ncol = 2,
  dimnames = list(student_id, c("quiz1", "quiz2"))
)

# list containing course, scores, and a named cutoffs vector
course_record <- list(
  course = "R Programming",
  scores = students,
  cutoffs = c(pass = 70, excellent = 90)
)

# Inspect the objects
typeof(section);      class(section);      length(section);      str(section)

## [1] "integer"
## [1] "factor"
```

```
## [1] 6
## Factor w/ 2 levels "A","B": 1 2 1 2 1 2
typeof(students); class(students); length(students); str(students); dim(students)

## [1] "list"
## [1] "data.frame"
## [1] 5
## 'data.frame': 6 obs. of 5 variables:
## $ student_id: chr "S01" "S02" "S03" "S04" ...
## $ section : Factor w/ 2 levels "A","B": 1 2 1 2 1 2
## $ quiz1 : num 82 91 76 88 95 69
## $ quiz2 : num 85 89 80 92 94 74
## $ passed : logi TRUE TRUE TRUE TRUE TRUE FALSE
## [1] 6 5
typeof(score_matrix); class(score_matrix); length(score_matrix); str(score_matrix); dim(score_matrix)

## [1] "double"
## [1] "matrix" "array"
## [1] 12
## num [1:6, 1:2] 82 91 76 88 95 69 85 89 80 92 ...
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:6] "S01" "S02" "S03" "S04" ...
## ..$ : chr [1:2] "quiz1" "quiz2"
## [1] 6 2
typeof(course_record); class(course_record); length(course_record); str(course_record)

## [1] "list"
## [1] "list"
## [1] 3
## List of 3
## $ course : chr "R Programming"
## $ scores : 'data.frame': 6 obs. of 5 variables:
## ..$ student_id: chr [1:6] "S01" "S02" "S03" "S04" ...
## ..$ section : Factor w/ 2 levels "A","B": 1 2 1 2 1 2
## ..$ quiz1 : num [1:6] 82 91 76 88 95 69
## ..$ quiz2 : num [1:6] 85 89 80 92 94 74
## ..$ passed : logi [1:6] TRUE TRUE TRUE TRUE TRUE FALSE
## $ cutoffs: Named num [1:2] 70 90
## ..- attr(*, "names")= chr [1:2] "pass" "excellent"
section <- factor(section, levels = c("A","B"))

students <- data.frame(
  student_id = student_id,
  section = section,
  quiz1 = quiz1,
  quiz2 = quiz2,
  passed = passed,
```

```

    stringAsFactors = FALSE
)

score_matrix <- matrix(
  c(quiz1, quiz2),
  nrow = 6, ncol = 2,
  dimnames = list(student_id, c("quiz1", "quiz2"))
)

course_record <- list(course = "R Programming",
                      score = students,
                      cutoffs = c(pass = 70, excellent = 90))

typeof(quiz1); class(quiz1); length(quiz1)

## [1] "double"
## [1] "numeric"
## [1] 6

typeof(section); class(section); length(section)

## [1] "integer"
## [1] "factor"
## [1] 6

typeof(score_matrix); class(score_matrix); dim(score_matrix)

## [1] "double"
## [1] "matrix" "array"
## [1] 6 2

typeof(students); class(students); dim(students)

## [1] "list"
## [1] "data.frame"
## [1] 6 6

typeof(course_record); class(course_record); length(course_record)

## [1] "list"
## [1] "list"
## [1] 3

# A vector contains elements of one atomic type
# List: Contain elements of different types and lengths
# Matrix: An atomic vectpr with a 2d dim attribute
# Data Frame: A list of equal-length vector

```

Part B

```
score_matrix["S04", "quiz2"]

## [1] 92
score_matrix[1:2, , drop = FALSE]

##      quiz1 quiz2
## S01      82    85
## S02      91    89
course_record["course"]

## $course
## [1] "R Programming"
course_record[["course"]]

## [1] "R Programming"
course_record$course

## [1] "R Programming"
# [ always preserves the structure. [[ extracts a single
# element and returns the element itself.
# $ returns a named element

students$average <- rowMeans(students[, c("quiz1", "quiz2")])

students$excellent <- students$average >= 90

# section A students with average >= 80
section_A_high <- students[students$section == "A" & students$average >= 80, ]

section_A_high[, c("student_id", "section", "average")]

##   student_id section average
## 1         S01      A    83.5
## 5         S05      A    94.5
avg_by_student <- students$average
names(avg_by_student) <- students$student_id
avg_by_student

##   S01 S02 S03 S04 S05 S06
## 83.5 90.0 78.0 90.0 94.5 71.5
```

Part C

```
students$average <- rowMeans(students[, c("quiz1", "quiz2")])

students$excellent <- students$average >= 90

section_A <- students[students$section == "A" & students$average >= 80, ]

section_A[, c("student_id", "section", "average")]
```

```
##   student_id section average
## 1         S01         A    83.5
## 5         S05         A    94.5

avg_by_student <- students$average
names(avg_by_student) <- students$student_id

# The calculations are vectorized because it is faster than element-by-element loops.
```

Problem 2

```
csv_text <- "sample_id,site,temp_c,ph,status
M01,North,18.2,7.1,ok
M02,South,20.5,,ok
M03,North,NA,6.8,review
M04,East,22.1,7.4,ok
M05,South,19.7,7.0,review
M06,East,23.0,NA,ok
M07,North,17.8,6.9,ok
M08,South,21.2,7.2,ok"

measurements <- read.csv(text = csv_text, na.strings = c("", "NA"),
                          stringsAsFactors = FALSE)

head(measurements)

##   sample_id  site temp_c  ph status
## 1      M01 North   18.2 7.1    ok
## 2      M02 South   20.5 NA    ok
## 3      M03 North    NA 6.8 review
## 4      M04  East   22.1 7.4    ok
## 5      M05 South   19.7 7.0 review
## 6      M06  East   23.0 NA    ok

str(measurements)

## 'data.frame':   8 obs. of  5 variables:
## $ sample_id: chr  "M01" "M02" "M03" "M04" ...
## $ site      : chr  "North" "South" "North" "East" ...
## $ temp_c    : num  18.2 20.5 NA 22.1 19.7 23 17.8 21.2
## $ ph       : num  7.1 NA 6.8 7.4 7 NA 6.9 7.2
## $ status    : chr  "ok" "ok" "review" "ok" ...

dim(measurements)

## [1] 8 5

names(measurements)

## [1] "sample_id" "site"      "temp_c"    "ph"        "status"

colSums(is.na(measurements))

## sample_id      site      temp_c      ph      status
##          0          0          1          2          0
```

```

measurements_complete <- measurements[complete.cases(measurements), ]

removed_ids <- measurements$sample_id[!complete.cases(measurements)]

# x == NA because NA is an unknown value so comparisons are unknown

```

Part B

```

measurements$site <- factor(measurements$site)
measurements$status <- factor(measurements$status)
levels(measurements$site)

## [1] "East" "North" "South"
levels(measurements$status)

## [1] "ok" "review"
measurements$temp_f <- measurements$temp_c * 9 / 5 + 32

measurements$ph_below_7 <- measurements$ph < 7

ns_ok <- measurements[
  complete.cases(measurements) &
  measurements$site %in% c("North", "South") &
  measurements$status == "ok",
]

ns_ok[, c("sample_id", "site", "temp_c", "temp_f", "ph")]

##   sample_id site temp_c temp_f ph
## 1      M01 North  18.2  64.76 7.1
## 7      M07 North  17.8  64.04 6.9
## 8      M08 South  21.2  70.16 7.2
mean(measurements$temp_c, na.rm = TRUE)

## [1] 20.35714
mean(measurements$temp_c[measurements$site == "South"], na.rm = TRUE)

## [1] 20.46667

```

Part C

```

A <- matrix(1:4, nrow = 2)
B <- matrix(5:8, nrow = 2)

A * B

##      [,1] [,2]
## [1,]    5   21
## [2,]   12   32

```

```
A %*% B
```

```
##      [,1] [,2]  
## [1,]   23  31  
## [2,]   34  46
```

```
dim(A * B)
```

```
## [1] 2 2
```

```
dim(A %*% B)
```

```
## [1] 2 2
```

*#Element-wise require that the matrices have the same dimernsions while matrix multiplication
does not require these constraints*

Problem 3

```
#Setup  
student_id <- paste0("P", sprintf("%02d", 1:8))  
scores <- c(95, 82, NA, 67, 74, 88, 59, 91)
```

```
# Part A  
grade_one <- function(  
  score,  
  a_min = 90,  
  b_min = 80,  
  c_min = 70,  
  d_min = 60  
) {  
  if (is.na(score)) {  
    return(NA_character_)  
  } else if (score >= a_min) {  
    return("A")  
  } else if (score >= b_min) {  
    return("B")  
  } else if (score >= c_min) {  
    return("C")  
  } else if (score >= d_min) {  
    return("D")  
  } else {  
    return("F")  
  }  
}
```

```
grade_one(NA)
```

```
## [1] NA
```

```
grade_one(90)
```

```
## [1] "A"
```

```
grade_one(80)
```

```
## [1] "B"
```

```
grade_one(85)
```

```
## [1] "B"
```

```
grade_one(74)
```

```
## [1] "C"
```

Part B

```
grades <- rep(NA_character_, length(scores))
for (i in seq_along(scores)) {
  grades[i] <- grade_one(scores[i])
}
names(grades) <- student_id
grades
```

```
## P01 P02 P03 P04 P05 P06 P07 P08
```

```
## "A" "B" NA "D" "C" "B" "F" "A"
```

```
# i represents the integer index position.
```

Part C-1

```
summarize_scores <- function(x, na.rm = TRUE, digits = 1) {
  c(
    count = length(x),
    missing = sum(is.na(x)),
    mean = round(mean(x, na.rm = na.rm), digits),
    sd = round(sd(x, na.rm = na.rm), digits),
    min = round(min(x, na.rm = na.rm), digits),
    max = round(max(x, na.rm = na.rm), digits)
  )
}
```

```
summarize_scores(scores)
```

```
##   count missing   mean    sd   min   max
##    8.0     1.0  79.4   13.3  59.0  95.0
```

```
summarize_scores(x = scores, na.rm = TRUE, digits = 2)
```

```
##   count missing   mean    sd   min   max
##    8.00     1.00  79.43  13.28  59.00  95.00
```

```
# Part C-2
```

```
plot_scores <- function(x, ...) {
  plot(seq_along(x), x, ...)
}
```

```
plot_scores(scores, type = "b", pch = 19, xlab = "Position", ylab = "Score", main = "Student Scores")
```

Student Scores

